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DEPARTMENT OF STATISTICS

Case Studies - Project 1

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Point Forecasts of COVID-19 New Cases and Deaths:
A Linear Modeling Approach

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Contents

1	Introduction	3
2	Problem and Data Description	3
2.1	Data Sources and Transformations	3
2.2	Raw v/s Log-Transformed Data	4
2.3	ACF and PACF Plots	5
3	Methods	5
3.1	Autoregressive Models	6
3.2	Lag Selection Using BIC	6
3.3	Autoregressive Models with Extra Variables	7
3.4	Software	7
3.5	Model Assumptions and Diagnostic Checks	8
4	Empirical Results	8
4.1	Diagnostic Test Results	9
4.1.1	Stationarity	9
4.1.2	Residual Checks	9
4.2	In-Sample Estimation Results	9
4.3	Seasonal Autoregressive Model Results	11
4.4	Out-of-Sample Forecast Results	13
4.5	Interpretation of Results	14
5	Conclusion	14
A	Appendix	16
A.1	Forecasts and Errors	16
A.2	Regime Indicator Check	17
A.3	Model Checks	18
	References	19

1 Introduction

COVID-19 had a strong effect on Germany. It affected hospitals, daily life, and the economy. Therefore, it is useful to study how COVID-19 cases and deaths changed over time and whether they can be forecasted. In this project, we forecast daily COVID-19 cases and deaths in Germany using time-series models. The data comes from the Robert Koch Institute (RKI) and covers the period from March 2020 to April 2026 [Robert Koch-Institut, 2026].

The main idea is to use past values of cases and deaths to predict future values. This is suitable for COVID-19 data because the numbers on one day are usually related to the numbers from previous days. The data also shows a weekly reporting pattern, because reported numbers are often lower on weekends and higher again at the start of the next week. For this reason, autoregressive and seasonal time-series models are useful for our project [Box and Jenkins, 1970, Hamilton, 1994].

We first look at the data using plots, the Autocorrelation Function (ACF), and the Partial Autocorrelation Function (PACF). After that, we estimate and compare different forecasting models. These include simple autoregressive models, models with additional variables such as age-group case numbers and temperature, and seasonal autoregressive models. Forecast performance is compared using out-of-sample forecast errors.

The results show that past values are important for forecasting cases and deaths. The weekly pattern is especially clear in the case data. Using more suitable lags improves the forecasts, while seasonal models give only small additional improvements.

The rest of the report is structured as follows. Section 2 describes the data. Section 3 explains the methods. Section 4 presents the results. Section 5 concludes the report.

2 Problem and Data Description

This section explains the data used in the report. The main data are daily COVID-19 cases and deaths in Germany. The data covers the period from March 2020 to April 2026. We also use age-group case numbers and temperature as extra variables.

2.1 Data Sources and Transformations

The daily case and death data come from the Robert Koch Institute (RKI). The RKI collects COVID-19 reports from local public health offices in Germany. These reports include confirmed infections and deaths [Robert Koch-Institut, 2026].

The case data are also given by age group. This is useful because infections can develop differently in different age groups. In this analysis, we use the age groups 0–4, 5–14, 15–34, 35–59, 60–79, and 80 years or older. Cases with unknown age classification are included as an additional regressor. These age-group case numbers are later used as extra variables in some

models.

Daily mean temperature is also used as an additional variable. The temperature data come from the Open-Meteo Historical Weather Archive API [Zippenfenig, 2026]. Since this report looks at Germany as a whole, Germany’s geographic centre, with coordinates 51.1657°N and 10.4515°E, is used as a simple approximation for the country’s temperature.

The raw case and death numbers have some very large peaks, especially during pandemic waves. To make these extreme values smaller, we use a log transformation [Box and Cox, 1964]. The transformation is given by:

$$y_t = \log(1 + \text{count}_t). \quad (1)$$

We add 1 before taking the logarithm so that days with zero deaths can also be used.

Table 2.1 shows the main summary statistics for the raw and log-transformed variables. The final dataset contains 2,193 daily observations.

Table 2.1: Summary statistics of daily COVID-19 cases and deaths.

Variable	Obs.	Mean	Std. Dev.	Min.	Q1	Median	Max.
Cases	2193	17840.63	41928.59	1.00	260.00	1455.00	307836.00
Deaths	2193	85.40	156.54	0.00	4.00	18.00	1300.00
Log cases	2193	7.41	2.56	0.69	5.56	7.28	12.64
Log deaths	2193	3.04	1.86	0.00	1.61	2.94	7.17

The raw case and death numbers are not evenly spread out. Most days have lower values, but some days have very high values. This can be seen from the large difference between the median and the maximum values. After using the log transformation, the extreme values become smaller, so the series is easier to analyse later.

2.2 Raw v/s Log-Transformed Data

Figure 2.1 shows the daily COVID-19 cases and deaths before and after the log transformation. This helps to compare the original data with the transformed data.

The raw case numbers have some very large peaks. The biggest peak is in early 2022, during the Omicron wave, when some days had more than 300,000 reported cases. After 2023, the reported case numbers become much lower. The death numbers are highest around the end of 2020 and the beginning of 2021, when daily deaths reached almost 1,300.

The log transformation makes the very large peaks smaller. This makes the smaller changes in the data easier to see. In the lower panels of Figure 2.1, the main waves are still visible, but the largest values do not dominate the graph as much. The weekly reporting pattern is also clearer. This pattern happens because fewer reports are usually processed on weekends, and some reports are added after the weekend.

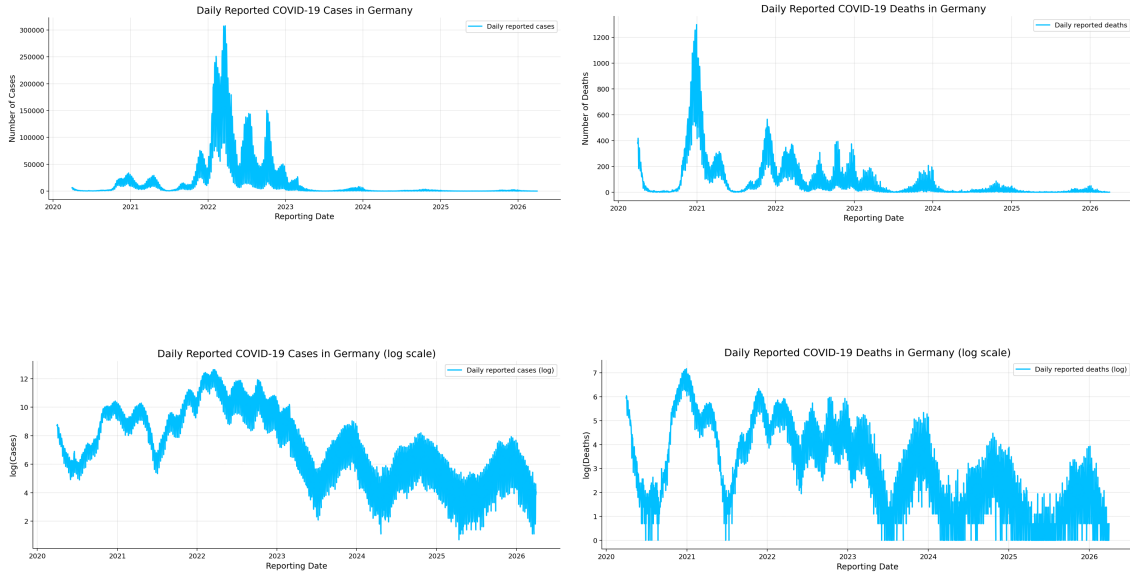


Figure 2.1: Raw daily COVID-19 cases and deaths compared with their log transformations for Germany from 2020 to 2026.

2.3 ACF and PACF Plots

To understand the patterns in the time series, we use ACF and PACF plots. These plots show how strongly current values are related to past values. Figure 2.2 shows the ACF and PACF plots for log cases and log deaths up to 40 lags.

The ACF plots show that both log cases and log deaths are strongly related to their past values. The correlations decrease slowly, so older values still matter. There are also clear spikes at lags 7, 14, 21, 28, and 35. This shows the weekly reporting pattern.

The PACF plots show which lags are most important. For log cases, lag 1 is the strongest. This means that yesterday's value is very important. There are also smaller spikes from lag 2 to lag 6 and another clear spike at lag 7, which again shows the weekly cycle.

For log deaths, the pattern is similar, but the effect of past values lasts a little longer. Lag 1 is still the strongest, and some other lags are important up to about one week. This may be because deaths are often reported with delay. Overall, the ACF and PACF plots show that autoregressive models are suitable for the data.

3 Methods

This section explains the models used to forecast daily COVID-19 cases and deaths in Germany. First, we use autoregressive models, which means that past values are used to predict future values. After that, we add extra variables, such as age-group case numbers and temperature. This section also explains how the lag length is chosen, which software is used, and how the models are checked.

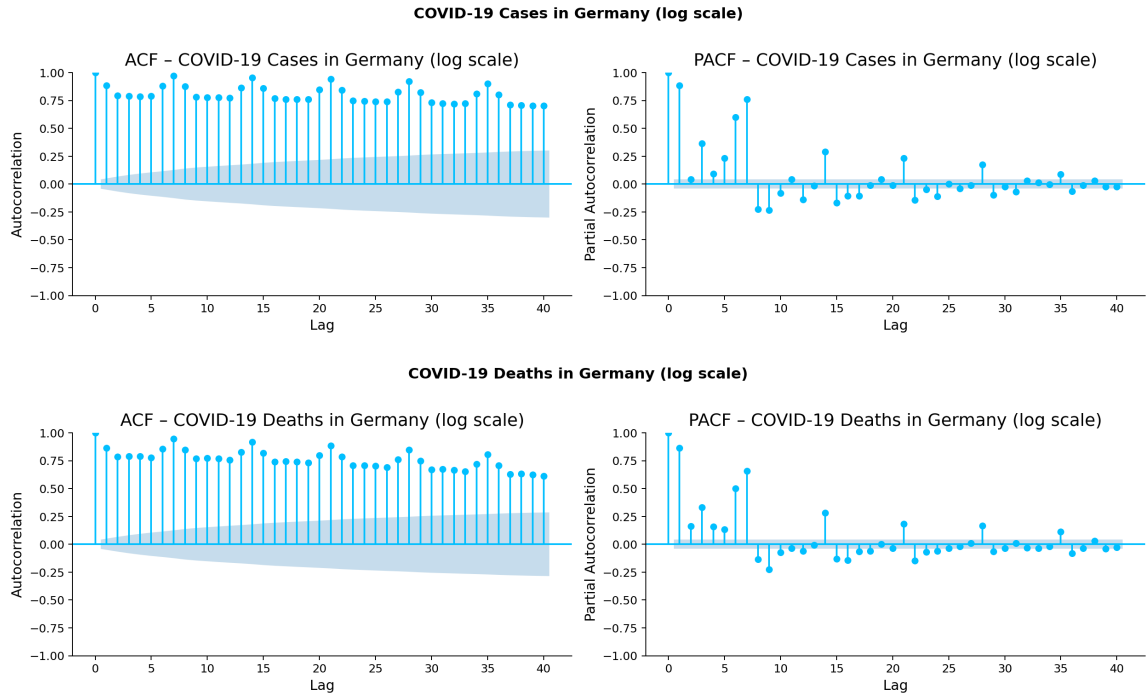


Figure 2.2: ACF and PACF plots for log COVID-19 cases and log COVID-19 deaths up to 40 lags.

3.1 Autoregressive Models

We first use autoregressive models. In these models, the current value is explained by past values of the same series. Let y_t be the log-transformed value on day t . For cases and deaths, it is defined as:

$$y_t = \log(1 + \text{count}_t).$$

An autoregressive model of order p , written as $AR(p)$, uses the past p values to explain the current value [Box and Jenkins, 1970, Hamilton, 1994]. The model is written as:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t. \quad (2)$$

Here, c is the constant, ϕ_i is the coefficient for lag i , and ε_t is the error term [Wooldridge, 2015].

The data have a strong weekly pattern. Because of this, an $AR(1)$ model may be too simple [Box and Jenkins, 1970, Hamilton, 1994]. We therefore, also use an $AR(7)$ model. This model uses the values from the last seven days. This helps the model include the weekly reporting pattern [Günther et al., 2021].

3.2 Lag Selection Using BIC

The data show a weekly pattern, so using 7 lags makes sense. However, the model should not use too few or too many lags. If too few lags are used, important past information may be

missed. If too many lags are used, the model may become too complicated and may fit random noise.

To choose the lag length in a more objective way, we use the Bayesian Information Criterion (BIC). The BIC helps select the lag length that gives a good fit without making the model too complex.

We estimate different autoregressive models with lag lengths from 1 up to a maximum value. For each lag length p , we calculate the Bayesian Information Criterion (BIC) [Schwarz, 1978]:

$$\text{BIC}(p) = -2 \ln(\hat{L}) + k \ln(T). \quad (3)$$

Here, $\hat{L}$ is the maximum likelihood value, k is the number of estimated parameters, and T is the number of observations. The best lag length is the one with the lowest BIC value.

The BIC is useful because it does not only look at how well the model fits the data. It also penalizes models with too many parameters. This helps us choose a model that is accurate but not too complicated [Schwarz, 1978].

3.3 Autoregressive Models with Extra Variables

In order to check whether extra information improves the forecasts, we also use ARX models. An ARX model uses past values of the target variable and additional variables. In this report, the additional variables are age-group infection numbers and temperature [Hamilton, 1994].

The ARX model is given by:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^m \beta_j X_{j,t} + \varepsilon_t. \quad (4)$$

Here, $X_{j,t}$ is the value of the additional variable j at time t , and β_j is its coefficient.

Two types of extra variables are used in the analysis. First, age-group infection numbers from the Robert Koch Institute are included. These variables help to check whether infections in different age groups give useful information for the forecast [Robert Koch-Institut, 2026]. Second, daily average temperature from the Open-Meteo API is included. This helps to check whether weather information improves the model [Zippenfenig, 2026]. A binary regime indicator for the post-April-2023 period is also included to account for structural changes in reporting behaviour. These variables are added to see whether they improve the forecasts compared with models that only use past values of cases or deaths.

3.4 Software

All analysis is done in Python. The data is cleaned and prepared using `pandas` and `numpy`. The time-series models are estimated with `statsmodels`. Weather data is downloaded from

the Open-Meteo API using `requests`. The figures and diagnostic plots are created using `matplotlib`.

3.5 Model Assumptions and Diagnostic Checks

After estimating the models, we check whether the residuals behave properly. This is important because the model results are more reliable when the errors do not show clear patterns [Wooldridge, 2015].

One important assumption is that the error term has an expected value of zero. This means that the model should not systematically overestimate or underestimate the values:

$$E(\varepsilon_t \mid y_{t-1}, y_{t-2}, \dots, X_t) = 0. \quad (5)$$

Another assumption is that the errors should have a constant variance over time. This means that the size of the errors should not become much larger or smaller in some periods. The errors should also not be correlated with each other over time.

If the residuals are still correlated, it means that the model has not captured all patterns in the data. This can make the standard errors unreliable and can make some coefficients look more important than they really are [Hamilton, 1994].

To check the model assumptions, we use different diagnostic tests. First, we use the Augmented Dickey-Fuller (ADF) test to check whether the series is stationary [Wooldridge, 2015]. After estimating the models, we use the Ljung-Box Q-test to check whether the residuals are still autocorrelated [Box and Jenkins, 1970].

The Ljung-Box test checks whether the residual autocorrelations up to lag H are jointly equal to zero. The test statistic is:

$$Q(H) = T(T+2) \sum_{k=1}^H \frac{\hat{\rho}_k^2}{T-k}. \quad (6)$$

Here, $\hat{\rho}_k$ is the residual correlation at lag k , and T is the number of observations used in the model. If the Ljung-Box test gives a low p -value, it means that the residuals are still correlated. This shows that the model has not captured all patterns in the data, so more lags may be needed.

We also look at residual plots. These plots help us see whether the errors stay stable over time. We also check the shape of the residual distribution. If the residuals are roughly symmetric and close to a normal distribution, the model is more suitable for forecasting.

4 Empirical Results

This section presents the results of our time-series models. First, we describe the main properties of the data. Then, we look at how well the models fit the past data. Finally, we compare

the forecasting performance using out-of-sample forecasts. The results show how important weekly reporting patterns are for COVID-19 cases and deaths.

4.1 Diagnostic Test Results

4.1.1 Stationarity

Before estimating the models, we check whether the series are stationary. The results are shown in Table 4.1. The ADF test shows that log cases are not stationary in levels, since the p-value is high. For log deaths, the result is stronger, but it is close to the 5% significance level.

After first differencing, both series become stationary. For log cases and log deaths, the results are clear, with p-values below 0.001. Since both first differences are stationary, the differenced specification is used for the SARIMA models

Table 4.1: Augmented Dickey-Fuller test results.

Series	ADF statistic	p-value	Result
Log cases (level)	-1.685	0.439	non-stationary
Log deaths (level)	-2.969	0.038	borderline
Δ log cases	-7.948	< 0.001	stationary
Δ log deaths	-7.084	< 0.001	stationary

4.1.2 Residual Checks

After estimating the models, we check whether the residuals still show autocorrelation. This is important because residuals should behave like random errors. We use the Ljung-Box test up to 40 lags. The residual ACF plots are shown in Figure 4.1.

Figure 4.1 shows clear differences between the models. The AR(1) model still has strong autocorrelation, especially at weekly lags. This means that the AR(1) model is too simple for the data. The AR(7) model reduces the autocorrelation, but some dependence still remains in the residual ACF plots.

The optimized AR models perform better. The optimized AR(50) model for cases and the optimized AR(36) model for deaths produce residuals that mostly stay within the 95% confidence bounds. This suggests that the longer lag models capture the time-series structure better.

4.2 In-Sample Estimation Results

Next, we compare how well the models fit the historical data. The main results are shown in Table 4.2. The table includes AIC, BIC, and RMSE on the log scale. These results help us compare which models fit the past data better.

The first results show that using more lags improves the model fit. For cases, the RMSE decreases from 1.1878 in the AR(1) model to 0.5157 in the AR(7) model. For deaths, the

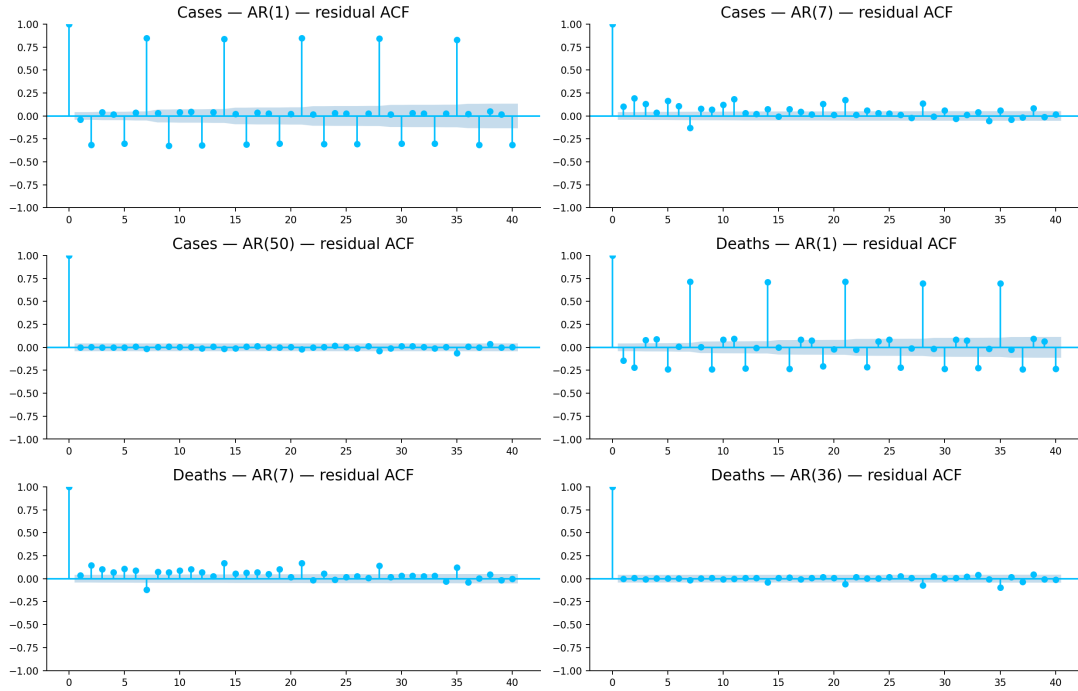


Figure 4.1: Residual ACF plots up to lag 40 for the AR(1), AR(7), and optimized AR models.

Table 4.2: In-sample model estimation and goodness-of-fit statistics.

Series	Model	AIC	BIC	RMSE (log)
Cases	AR(1)	6981.1	6998.2	1.1878
Cases	AR(7)	3326.6	3377.8	0.5157
Cases	AR(50)	2297.9	2592.7	0.4037
Cases	ARX(7)+regime	3167.4	3269.7	0.4975
Deaths	AR(1)	5922.6	5939.7	0.9330
Deaths	AR(7)	3412.7	3463.9	0.5260
Deaths	AR(36)	2744.6	2960.3	0.4492
Deaths	ARX(7)+regime	3239.5	3341.8	0.5059

RMSE decreases from 0.9330 to 0.5260. This shows that the AR(7) model fits the data better than the AR(1) model.

The BIC search shows that the best lag length is 50 lags for cases and 36 lags for deaths. These lag lengths give a good balance between model fit and model complexity. Figure 4.2 shows how the AIC and BIC values change for different lag lengths.

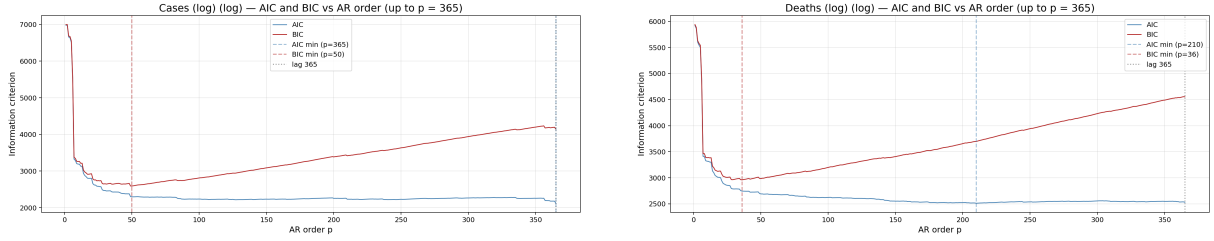


Figure 4.2: AIC and BIC as a function of AR order $p = 1, \dots, 365$ for the log-transformed case and death series.

To check the basic model performance, we calculate in-sample predictions and RMSE values. These errors are calculated using the same data that were used to estimate the models. Therefore, the results can look better than the real forecasting performance.

The in-sample results are still useful for a first comparison. However, the more important test is the out-of-sample forecast evaluation in Section 4.4. Figure 4.3 shows how the fitted values compare with the actual log-transformed values. All evaluation is done on the logarithmic scale.

4.3 Seasonal Autoregressive Model Results

To check for seasonal patterns, we also estimate seasonal autoregressive models. These models are compared with the AR(7) model. The results are shown in Table 4.3.

Table 4.3: In-sample RMSE on the log scale for SARIMA models compared with the AR(7) baseline.

Target	Model	RMSE (log)
Cases	AR(7) baseline	0.5157
Cases	SARIMA(7, 1, 0)(1, 0, 0) ₃₆₅	0.4897
Cases	SARIMA(50, 1, 0)(1, 0, 0) ₃₆₅	0.3939
Deaths	AR(7) baseline	0.5260
Deaths	SARIMA(7, 1, 0)(1, 0, 0) ₃₆₅	0.5034
Deaths	SARIMA(36, 1, 0)(1, 0, 0) ₃₆₅	0.4502

The seasonal models perform better than the standard AR(7) model. For cases, the in-sample RMSE decreases from 0.5157 in the AR(7) model to 0.4897 in the SARIMA(7) model.

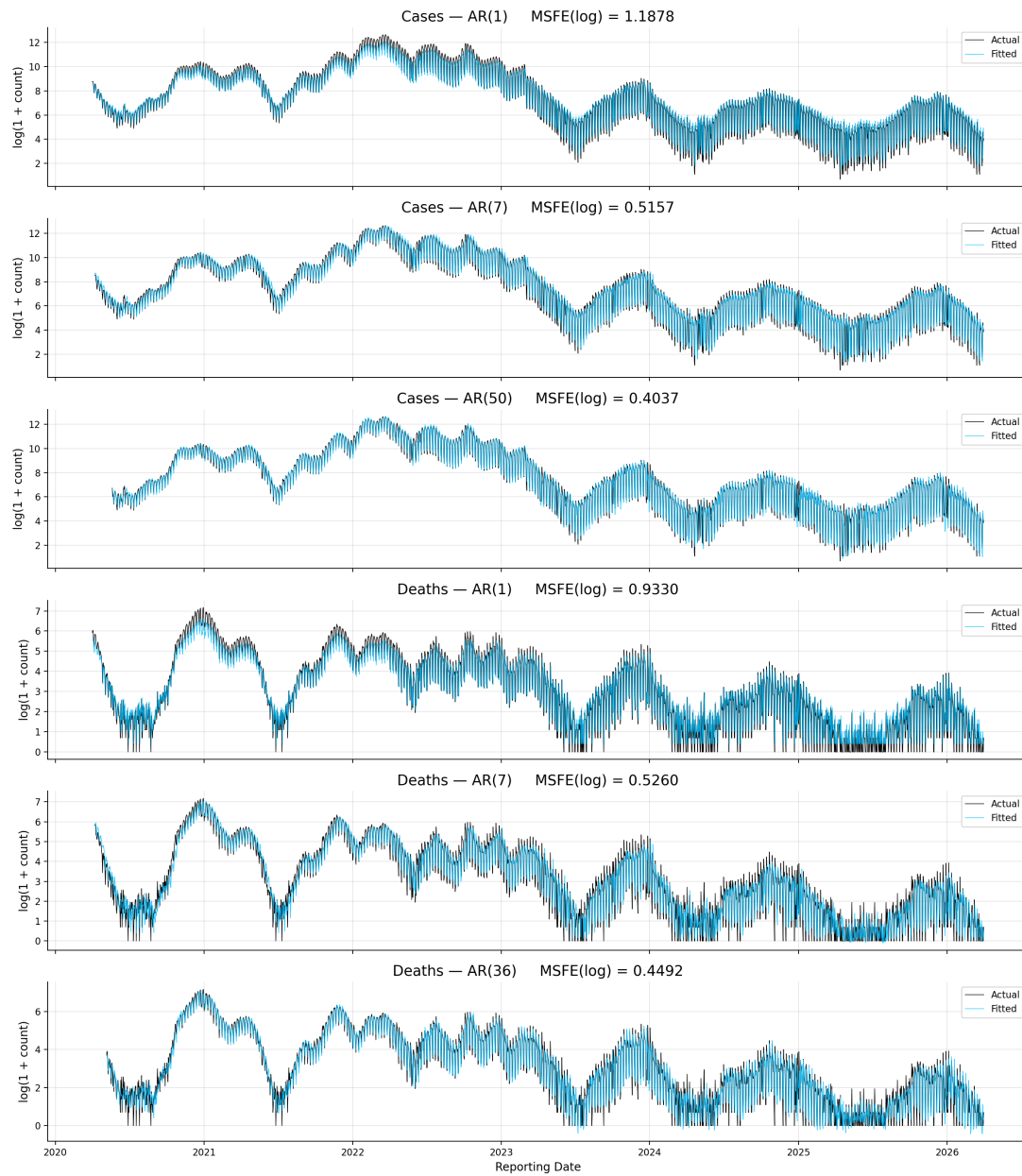


Figure 4.3: In-sample fitted values compared with actual log-transformed observations.

For deaths, it decreases from 0.5260 to 0.5034. This shows that adding seasonal information improves the model fit.

The optimized seasonal models perform even better. For cases, the SARIMA(50) model has the lowest in-sample RMSE, with a value of 0.3939. For deaths, the SARIMA(36) model has an in-sample RMSE of 0.4502. These results show that the optimized seasonal models fit the historical data best.

4.4 Out-of-Sample Forecast Results

Table 4.4 shows the rolling out-of-sample forecast results. The models are estimated using a rolling window of 1,000 days. Forecast performance is compared using RMSFE and Theil's U .

Table 4.4: Out-of-sample rolling-window results on the log scale. Training window: 1,000 days.

Target	Model	OOS RMSFE (log)	Theil's U
Cases	AR(1)	1.5425	0.9367
Cases	AR(7)	0.6714	0.4077
Cases	AR(50)	0.5603	0.3402
Cases	AR(7)+exog	0.6426	0.3902
Cases	SARIMA(7)	0.6122	0.3717
Cases	SARIMA(50)	0.5482	0.3329
Deaths	AR(1)	1.0917	0.9363
Deaths	AR(7)	0.6165	0.5288
Deaths	AR(36)	0.5609	0.4810
Deaths	AR(7)+exog	0.5828	0.4999
Deaths	SARIMA(7)	0.5860	0.5026
Deaths	SARIMA(36)	0.5641	0.4838

The AR(1) model performs the worst. Its Theil's U value is close to 1 for both cases and deaths. This means that it is only slightly better than the naive benchmark.

The AR(7) model performs much better. For cases, the RMSFE decreases from 1.5425 in the AR(1) model to 0.6714 in the AR(7) model. This is an improvement of about 56%. For deaths, the RMSFE decreases from 1.0917 to 0.6165, which is an improvement of about 44%. This shows that the weekly lag structure is important for forecasting.

The optimized models improve the results further. For cases, the AR(50) model has an RMSFE of 0.5603. The best case forecast comes from the SARIMA(50) model, with an RMSFE of 0.5482. For deaths, the best model is AR(36), with an RMSFE of 0.5609. The SARIMA(36) model is very close, with an RMSFE of 0.5641.

To check whether the forecast differences are statistically significant, we use the Diebold-Mariano test [Diebold and Mariano, 1995]. Compared with AR(1), all larger models perform

significantly better. The optimized AR models also perform better than AR(7). For cases, AR(50) beats AR(7) with $DM = 4.09$ and $p < 0.001$. For deaths, AR(36) beats AR(7) with $DM = 3.96$ and $p < 0.001$.

The difference between the optimized AR and SARIMA models is small. For cases, SARIMA(50) is slightly better than AR(50), but the difference is not statistically significant ($DM = 0.51$, $p = 0.61$). For deaths, AR(36) is slightly better than SARIMA(36), but this difference is also not statistically significant ($DM = -0.46$, $p = 0.65$). Overall, the optimized lag structure is more important than adding an extra seasonal component.

4.5 Interpretation of Results

The results show that weekly reporting patterns are very important in the COVID-19 data. This can be seen from the ACF plots, the residual checks, and the better performance of models with more lags. The AR(7) model already improves strongly compared with AR(1), because it captures the weekly cycle in the data.

The optimized AR models perform even better. For cases, the best pure AR model uses 50 lags, while for deaths it uses 36 lags. This shows that the data contain dependence beyond only one week. The longer lag structure helps the model capture more of the time-series pattern.

The ARX models show that age-group case numbers, temperature, and a post-April-2023 regime indicator can add some useful information. However, their improvement is smaller than the improvement from using a better lag structure. This suggests that past values of cases and deaths are the most important predictors in this analysis.

The seasonal models also help, especially for cases. However, the Diebold-Mariano tests show that the optimized AR and SARIMA models are not significantly different from each other. Overall, the results suggest that weekly reporting patterns and longer autoregressive lag structures are the main drivers of forecast accuracy.

5 Conclusion

This report examined which linear time-series models can forecast daily COVID-19 cases and deaths in Germany most accurately. The models were compared using forecast errors on the log scale. We used autoregressive models, models with extra variables, and seasonal models.

The results were evaluated in two ways. First, we checked how well the models fit the historical data. Second, we tested their forecasting performance with rolling out-of-sample forecasts. The Diebold-Mariano test was also used to compare whether the differences between models were statistically important [Diebold and Mariano, 1995].

The analysis shows that the data have a strong weekly pattern. This is mainly caused by the reporting cycle. The BIC results show that longer lag structures fit the data better. The best lag

length is 50 lags for cases and 36 lags for deaths. This means that the models need more than only one week of past values.

The ARX(7) model also uses extra variables, such as age-group infection numbers, lagged temperature, and a post-April-2023 regime indicator. These variables improve the forecasts slightly. However, the improvement is smaller than the improvement from using the optimized AR models.

In the rolling out-of-sample forecasts, the SARIMA(50) model performs best for cases. It has an RMSFE of 0.5482 and a Theil's U value of 0.333. For deaths, the AR(36) model performs best, with an RMSFE of 0.5609 and a Theil's U value of 0.481 [Theil, 1966].

The Diebold-Mariano tests show that the differences between the optimized AR and SARIMA models are not statistically significant [Diebold and Mariano, 1995]. This means that, after the lag structure is optimized, adding the seasonal component does not clearly improve the forecasts.

The results also show that the extra variables, such as temperature and age-group infection numbers, help only a little. Overall, the optimized autoregressive models are more important for forecasting than the extra variables.

Future research could improve the models in several ways. First, multivariate time-series models such as VAR models could be used. These models would allow cases, deaths, and other variables to be analysed together instead of separately [Hamilton, 1994, Stock and Watson, 2001]. Second, more flexible models could be used. For example, state-space models, structural break models, or Markov-switching models may be better at capturing changes in the pandemic and in reporting behaviour over time [Hamilton, 1989, Bai and Perron, 2003, Wood, 2017]. Future studies could also include more explanatory variables, such as mobility data, wastewater data, or other indicators of disease activity. Regularization methods such as LASSO could also help choose the most important variables and lags [Tibshirani, 1996]. Overall, these extensions may improve forecast performance, especially during periods with rapid changes and high uncertainty [Timmermann, 2006, Gneiting and Raftery, 2007].

A Appendix

This appendix presents additional results and diagnostic checks that support the findings reported in Section 4.

A.1 Forecasts and Errors

Figure A.1 compares the out-of-sample forecasts with the observed values over the rolling evaluation period for both cases and deaths.

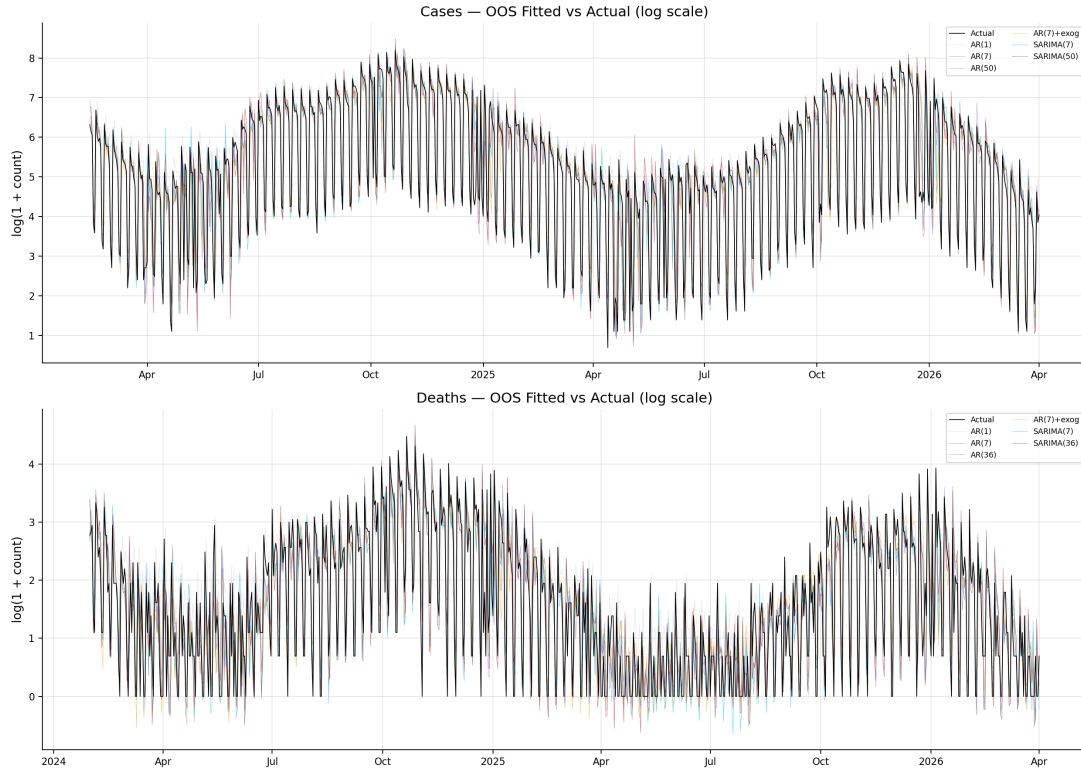


Figure A.1: Out-of-sample forecasts and observed values over the rolling forecast period.

To examine forecast errors in more detail, Figure A.2 shows the daily forecast errors over the evaluation period. Figure A.3 presents the cumulative squared errors and rolling RMSFE values for all model specifications, allowing a comparison of forecasting performance over time.

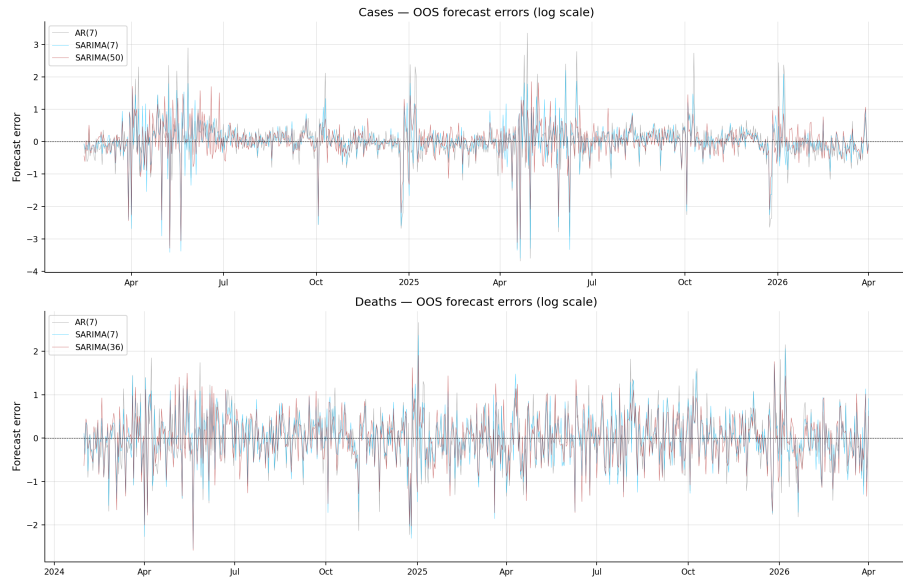


Figure A.2: Daily forecast errors over the rolling forecast period.

A.2 Regime Indicator Check

Table A.1 shows what happens when we add a regime indicator for the period after April 2023. This variable is added to the best extended AR(7) model. The results show only very small improvements.

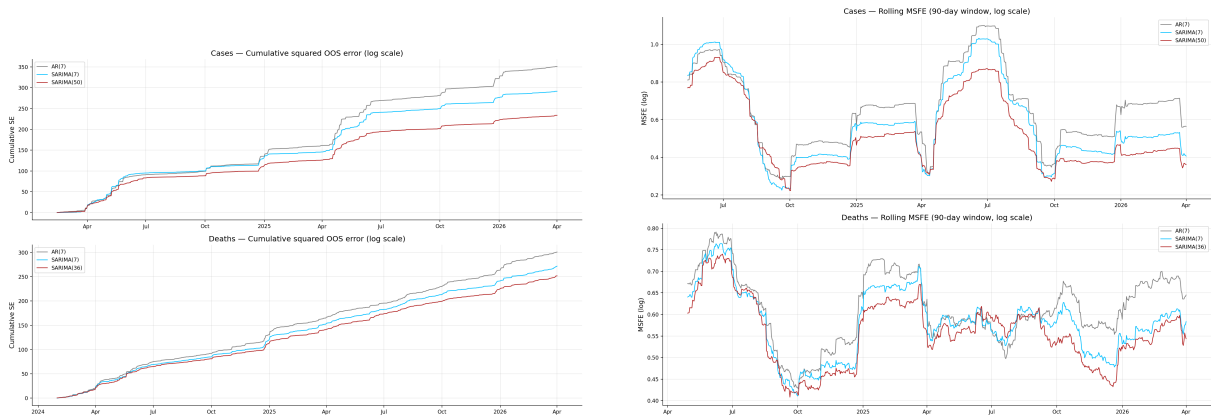


Figure A.3: Cumulative squared forecast errors and rolling RMSFE values over the evaluation period.

Table A.1: Robustness check: effect of including a binary regime indicator (post April 2023) in the best extended AR(7) specification.

Series	Model	AIC	BIC	RMSE (log)
Cases	AR(7) + age(14) + temp(7) + regime	3167.4	3269.7	0.4975
Cases	AR(7) + age(14) + temp(7)	3196.0	3292.6	0.5011
Deaths	AR(7) + age(14) + temp(7) + regime	3239.5	3341.8	0.5059
Deaths	AR(7) + age(14) + temp(7)	3268.0	3364.6	0.5094

A.3 Model Checks

Figure A.4 and Figure A.5 show the periodogram and residual diagnostic plots. These figures help us check whether the final models are suitable.

The periodogram shows the main repeating patterns in the data. The residual plots show whether there is still autocorrelation left after estimating the models. The ACF and PACF plots show that most residual autocorrelations stay inside the confidence bounds. This means that little serial dependence remains after the models are estimated.

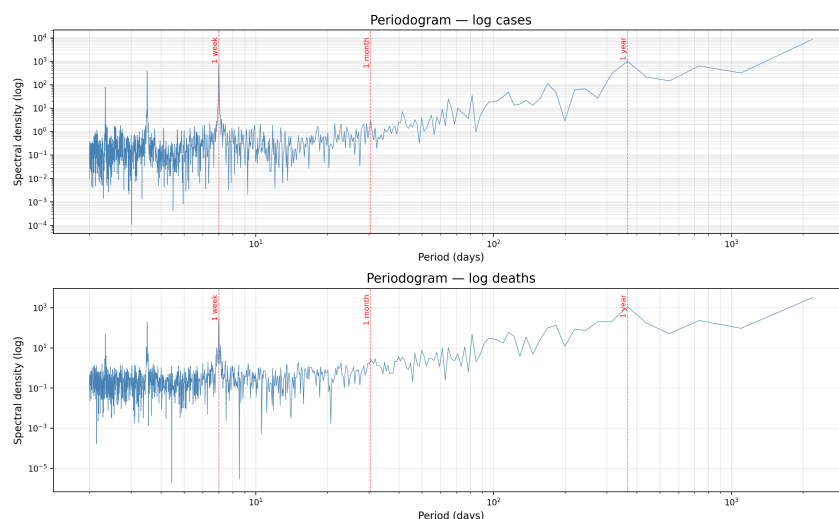


Figure A.4: Spectral density (Periodogram) analysis identifying dominant seasonal cycles.

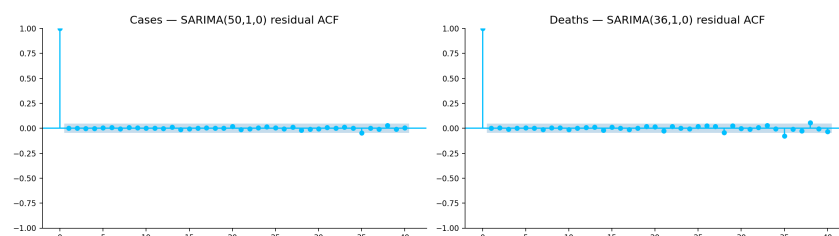


Figure A.5: Residual ACF plots for the optimal model specifications.

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